UNIVERSITY OF ENGINEERING & TECHNOLOGY MARDAN

COMPUTER SCIENCE

CS-305L: Parallel & Distributed Computing Lab

SUBMITTED BY

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REG NO : 23MDBCS 435

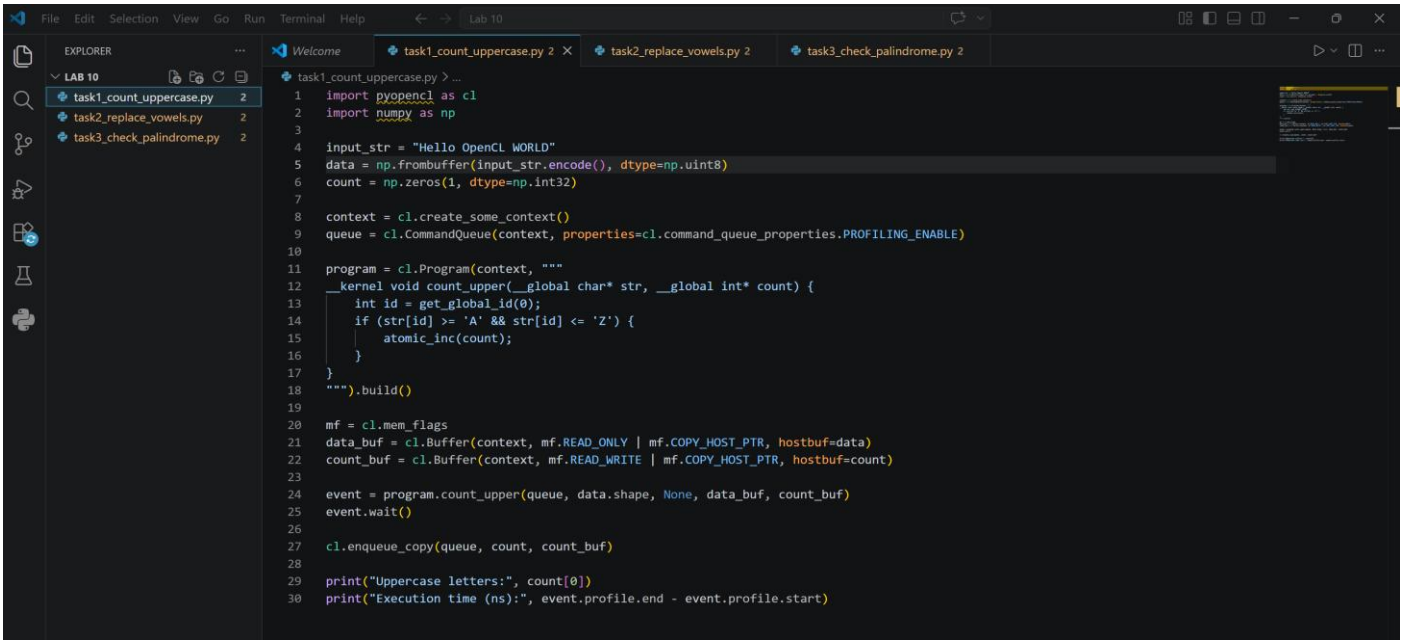
SECTION : CS-A

SEMESTER : 6th

Date: 22/4/2026

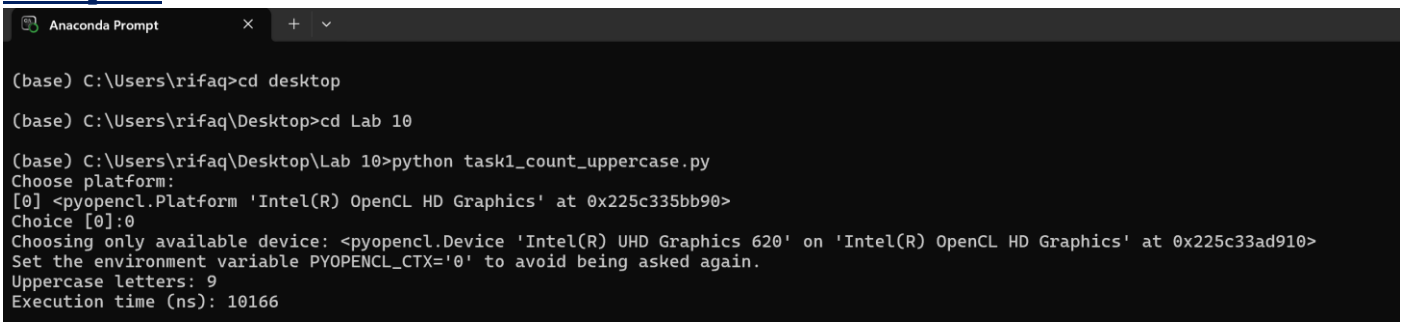
Lab 10

Task 1



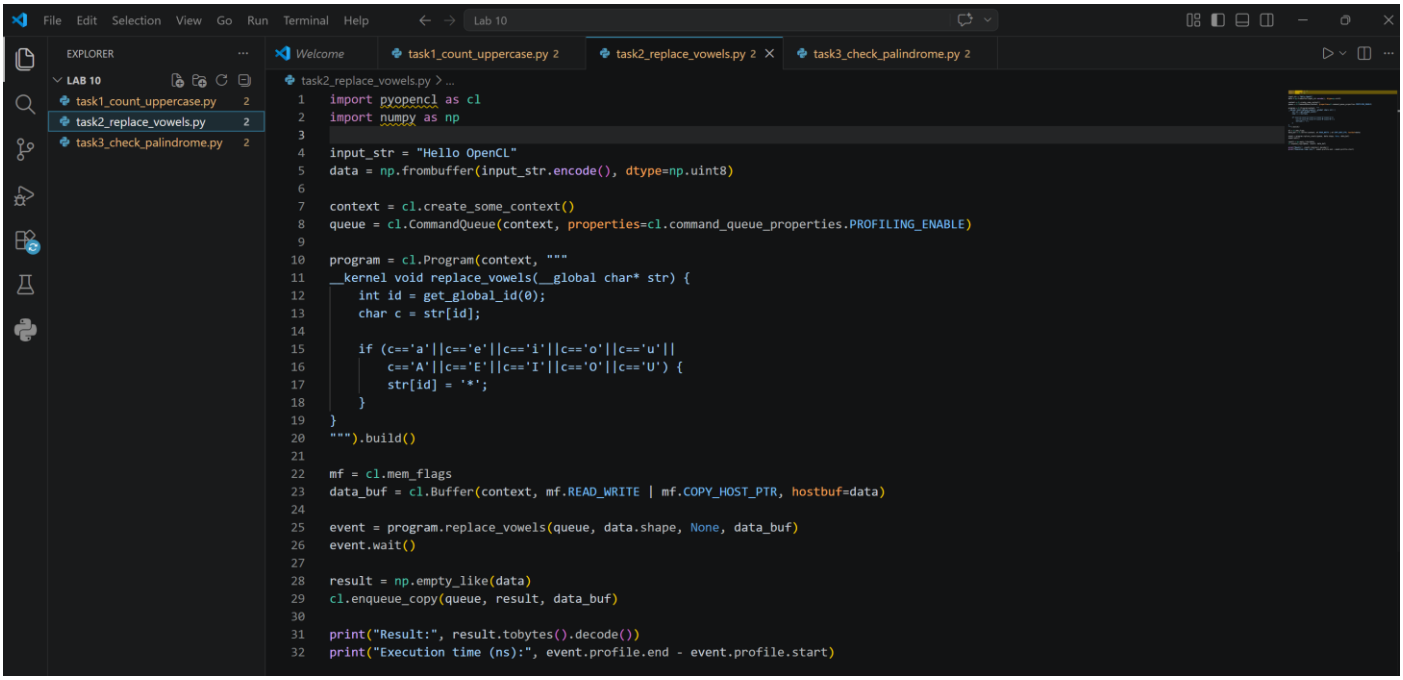
```python
import pyopencl as cl
import numpy as np

input_str = "Hello OpenCL WORLD"
data = np.frombuffer(input_str.encode(), dtype=np.uint8)
count = np.zeros(1, dtype=np.int32)

context = cl.create_some_context()
queue = cl.CommandQueue(context, properties=cl.command_queue_properties.PROFILING_ENABLE)

program = cl.Program(context, """
__kernel void count_upper(__global char* str, __global int* count) {
    int id = get_global_id(0);
    if (str[id] >= 'A' && str[id] <= 'Z') {
        atomic_inc(count);
    }
}
""").build()

mf = cl.mem_flags
data_buf = cl.Buffer(context, mf.READ_ONLY | mf.COPY_HOST_PTR, hostbuf=data)
count_buf = cl.Buffer(context, mf.READ_WRITE | mf.COPY_HOST_PTR, hostbuf=count)

event = program.count_upper(queue, data.shape, None, data_buf, count_buf)
event.wait()

cl.enqueue_copy(queue, count, count_buf)

print("Uppercase letters:", count[0])
print("Execution time (ns):", event.profile.end - event.profile.start)
```

Output



```
Anaconda Prompt

(base) C:\Users\rifaq>cd desktop

(base) C:\Users\rifaq\Desktop>cd Lab 10

(base) C:\Users\rifaq\Desktop\Lab 10>python task1_count_uppercase.py
Choose platform:
[0] <pyopencl.Platform 'Intel(R) OpenCL HD Graphics' at 0x225c335bb90>
Choice [0]:0
Choosing only available device: <pyopencl.Device 'Intel(R) UHD Graphics 620' on 'Intel(R) OpenCL HD Graphics' at 0x225c33ad910>
Set the environment variable PYOPENCL_CTX='0' to avoid being asked again.
Uppercase letters: 9
Execution time (ns): 10166
```

Task 2

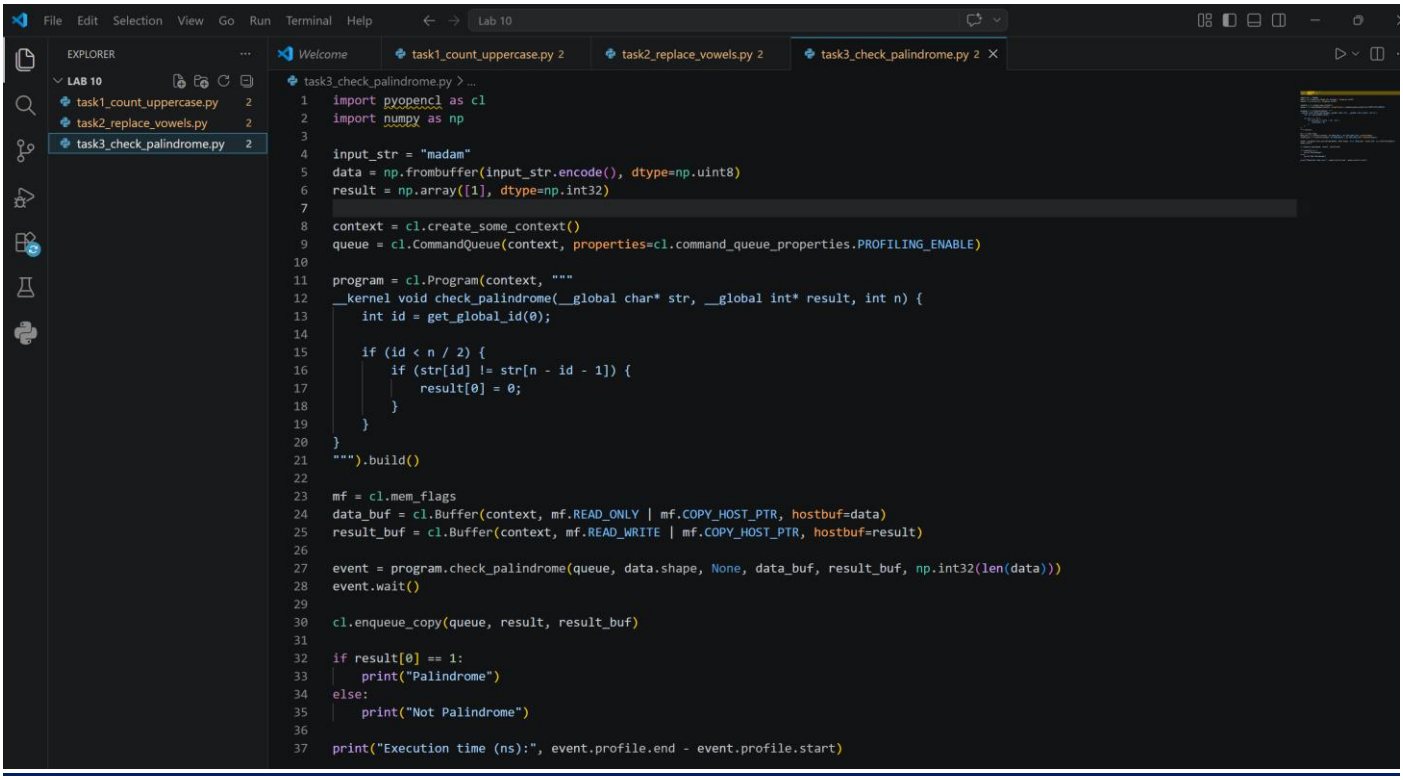


```python
import pyopencl as cl
import numpy as np

input_str = "Hello OpenCL"
data = np.frombuffer(input_str.encode(), dtype=np.uint8)

context = cl.create_some_context()
queue = cl.CommandQueue(context, properties=cl.command_queue_properties.PROFILING_ENABLE)

program = cl.Program(context, """
__kernel void replace_vowels(__global char* str) {
    int id = get_global_id(0);
    char c = str[id];

    if (c=='a'||c=='e'||c=='i'||c=='o'||c=='u'||
        c=='A'||c=='E'||c=='I'||c=='O'||c=='U') {
        str[id] = '*';
    }
}
""").build()

mf = cl.mem_flags
data_buf = cl.Buffer(context, mf.READ_WRITE | mf.COPY_HOST_PTR, hostbuf=data)

event = program.replace_vowels(queue, data.shape, None, data_buf)
event.wait()

result = np.empty_like(data)
cl.enqueue_copy(queue, result, data_buf)

print("Result:", result.tobytes().decode())
print("Execution time (ns):", event.profile.end - event.profile.start)
```

Output

```
(base) C:\Users\rifaq\Desktop\Lab 10>python task2_replace_vowels.py
Choose platform:
[0] <pyopencl.Platform 'Intel(R) OpenCL HD Graphics' at 0x1aa00115390>
Choice [0]:0
Choosing only available device: <pyopencl.Device 'Intel(R) UHD Graphics 620' on 'Intel(R) OpenCL HD Graphics' at 0x1aa00148b10>
Set the environment variable PYOPENCL_CTX='0' to avoid being asked again.
Result: H*ll* *p*nCL
Execution time (ns): 13666

(base) C:\Users\rifaq\Desktop\Lab 10>
```

Task 3



Output

```
(base) C:\Users\rifaq\Desktop\Lab 10>python task3_check_palindrome.py
Choose platform:
[0] <pyopencl.Platform 'Intel(R) OpenCL HD Graphics' at 0x25fc7aaee90>
Choice [0]:0
Choosing only available device: <pyopencl.Device 'Intel(R) UHD Graphics 620' on 'Intel(R) OpenCL HD Graphics' at 0x25fc7ad2890>
Set the environment variable PYOPENCL_CTX='0' to avoid being asked again.
Palindrome
Execution time (ns): 9499

(base) C:\Users\rifaq\Desktop\Lab 10>
```